

Enshan Chen

🏠 Postdoc, PhD

CONTACT

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Employment

Postdoc, on Digital Twins
TU Delft, Civil Engineering
(QS ranking **5th globally**)
2025-2026

EDUCATION

Master of Computer Science
Georgia Tech
2025–2027

PhD in Urbanism
(QS ranking **3rd globally**)
TU Delft
2021–2026

Master of Urban Design
Tongji University
2018–2021

Bachelor of Architecture
Chongqing University
2011–2016

SKILLS

Research skills: *Statistical analysis and modeling* (linear regression, spatial regression, discrete choice modeling, etc.), *qualitative research methods*, *quantitative research methods*, *research design*

Programming: Python, R, C#, HTML+CSS, LaTeX, Rhinoscript, etc. I intensively do *vibe coding* and are increasingly capable of more programming languages. **Proficient software:** (*Analytics*) QGIS, MassMotion, SPSS; SUMO; etc. (*Design*) Rhino, Grasshopper, Unreal Engine; etc. *ArchicAD*; (*Rendering*) V-Ray, Enscape; Photoshop, Illustrator; Figma; etc. I am a *faster learner* of software.

RESEARCH / PROJECT EXPERIENCE

POSTDOC: In *XCarCity* project, I develop: 2025-2026

- Federated *digital twins* for assessing multi-sensory user experience, based on *simulations* of traffic, visual perception, sunlight, wind, noise, etc.; a *genetic algorithm* is used for model calibration; I also build *web* interface to visualize the results.
- A macroscopic *pedestrian modeling* framework, predicting pedestrian flow using information of buildings and travel surveys. Technical aspects involved include: data scrapping, *gravity models*, *normalization* methods, etc. I *optimized the model to run 100 times faster*, by using CUDA (*GPU acceleration*) and the supercomputer of TU Delft.

PHD: Addressing overcrowding and underutilization in railway station areas: problems, assessment, and design 2021-2025

- Built an *assessment framework* for assessing *user experience* in station areas, based on *simulations* of pedestrian movement and *visual experience*; Published related papers in Q1 journals (Click [paper 1](#) and [paper 2](#); and as a software plugin – [PandaAnalytics](#)).
- Simulated *pedestrian movement* using algorithms including *social forces* and *Dijkstra's algorithm*. Calibrated it through a *genetic algorithm*. Incorporated a *feedback control loop*.
- Built *machine learning models* (incl. linear regression, neural networks of classification and regression) for pedestrian visual simulation.
- Extracting features from street images by *image segmentation* and manual annotation.
- Removed *spatial autocorrelation* in data by conducting spatial regression.
- Automated pedestrian counting, by deploying *object detection and tracking* algorithms – YOLO and Deepsort, utilizing the supercomputer DelftBlue.
- Others: Published two more software plugins [PandaModeling](#) (for using *LLMs* to create Rhino models) and [PandaMeasuring](#) (for analyzing urban blocks' density); Built [a website](#) to show design principles for railway stations.

MASTER'S (Computer Science) Small projects within courses: 2025-2026

- A small *trading system* using algorithms like KNN, decision trees, *reinforcement learning*, bagging, etc.; indicators like Bollinger Bands, Sharpe Ratio, RSI, moving average, etc.; manage and evaluate portfolios (in [Machine Learning for Trading](#))
- A project of *robot localization*, route planning, and control—Location using histogram filter, kalman filter, particle filter, and SLAM; Route planning using DFS, BFS, and A star; Control using bicycle model, and PID (in [Robotics: AI Techniques](#))
- Dealing with chess movement, using algorithms like Simulated Annealing, Random Forests; using knowledge like *Bayesian networks* (in [Artificial Intelligence](#))

MASTER'S (Urban Design): Urban form of railway station areas 2018-2021

- Extracted data of 50+ stations from Open Street Map using QGIS; analyzed the density and function diversity; published as [a journal article](#).
- Analyzed visibility (Isovist) in the Utrecht Central Station area, by coding in Python and Grasshopper; published as [a journal article](#).

RESEARCH INTERESTS

- Model, simulate, and design complex systems; Incorporate AI to analyze user experience/behavior/psychology, and to automate and generation solutions;

- Relevant technologies: LLMs (for analytics and general purposes); Diffusion model (for generating synthetic data); agent-based simulation, active inference, computer vision, machine/deep learning (for simulation); VR, biometrics (for data collection) Reinforcement learning, generative AI (for computational design)

MISCELLANEOUS

- **Working experiences:** 2016-2017, Shanghai Tianhua Architectural Design Co., architect; 2018-2021 (part-time), Tongji Architectural Design and Research Institute, urban designer. I have won multiple awards in architectural design competitions.
- **Design portfolios** during [the master's degree](#), and [the bachelor's degree](#) time.
- For a full list of **publications**, see my [CV](#).

Enshan Chen – Curriculum Vitae

(As of July 2026; Please review the digital version to access embedded links.)

1 Personal information

Enshan Chen

Postdoc, working on **digital twins**, at Delft University of Technology.

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Personal website: <https://1309928130.github.io/>

2 Education

Delft University of Technology (TU Delft)

(2025 QS Ranking in Architecture: No.3)

Ph.D., Urbanism,

2021-2015

Thesis: *Towards flexible uses of railway station areas that promote station-city integration: Spatial relevance, design, and assessment.*

This PhD project involves intensive computational analytics, as well as research-by-design. It is a transdisciplinary research across *urban+architectural design, computer science, crowd management, and environmental psychology*. The content related to technology is listed below:

- Built an *assessment framework* for assessing **user experience** in station areas, based on *simulations* of pedestrian movement and visual experience; Published it in two Q1 journals (see my *Publications*; and as a software plugin – *PandaAnalytics*.
- Simulated **pedestrian movement** using algorithms including *social forces* and *Dijkstra's* algorithm.
- Built **machine learning models** (incl. linear regression, neural networks of classification and regression) for pedestrian visual simulation.
- Extracting features from street images by **image segmentation** and manual annotation.
- Removed **spatial autocorrelation** in data by conducting spatial regression.
- Automated pedestrian counting, by deploying **object detection and tracking** algorithms – YOLO and Deepsort, utilizing the supercomputer DelftBlue.
- Others: Published two more software plugins *PandaModeling* (for using LLMs to create Rhino models) and *PandaMeasuring* (for analyzing urban blocks' density); Built *a website* to show design principles for railway stations.

Georgia Institute of Technology

Master of Computer Science

2025-2027

Tongji University

(2025 QS Ranking in Architecture: No.11)

Master of Architecture (Urban Design),

2018-2021

Thesis: *Urban form of station areas in the context of station-city integration*

Chongqing University

Bachelor of Architecture,

2011-2016

3 Working History

11/2025-10/2026, **Postdoc**, The Faculty of Civil Engineering and Geosciences, TU Delft. Delft, The Netherlands

- Develop (1) *data-driven* federated **digital twins** assessing **multi-sensory user experience** in urban districts, (2) a macroscopic **pedestrian model** predicting pedestrian flow; in project [XCarCity](#).

03/2018-09/2021, **Urban designer** (Part-time), Tongji Architectural Design (group) Co.,Ltd. Shanghai, China

- Urban Design of Core Area In Zhengzhou South Railway Station
- Urban Design Of West Lake Area In Zhangzhou

07/2016-07/2017, **Architect**, Shanghai TIANHUA Architecture Planning & Engineering Ltd. Shanghai, China

- Beijing Mentougou Longfor Shopping Mall
- Taiyuan Yangjiayu Macalline Residential District Project
- Partially participated in 10+ other projects

4 Certificate

First-class Architect License (China) (一级注册建筑师), 2023 - Present

Obtained through a rigorous certification process involving **nine exams**, covering both knowledge-based and drawing-based questions across a range of architectural fields, including:

Knowledge-Based Exams:

- Architectural Design
- Building Materials and Construction Detailing
- Building Economics, Construction, and Design Management
- Pre-Design and Site Design
- Building Structures
- Building Physics and Building Equipment

Drawing-Based Exams:

- Architectural Scheme Design
- Site Design (Masterplan)
- Architectural Technical Design

5 Technical skills

Languages: Chinese, English

Programming: Python, C#, R, HTML+CSS

Software skills: (*3D-modeling, computational design, BIM*) Rhino, Grasshopper, SketchUp, AutoCAD, ArchiCAD; (*3D-rendering*) V-ray, Lumion, Inscape; (*Graphic-design, UX*) Photoshop, InDesign, Illustrator, Figma; (*Simulation, urban analytics, statistics, etc.*) QGIS, MassMotion, SPSS, sDNA, UNA

I developed a few **software programs**:



PandaModeling –A Rhino plugin that uses large language models to build 3D models. *[Written in C#]*



PandaMeasuring –A Rhino plugin that automates the measuring of floor areas, useful for mapping density and urban form. *[Written in Grasshopper; Based on part of my Master's degree work]*



PandaAnalytics –A Rhino plugin for assessing user experience in station areas. *[Written in Python + C#; Based on part of my PhD research work]*

Other **research skills**:

- Statistical analysis,
- Quantitative research methods,
- Qualitative research methods

6 Design

Design competition winnings

(Student period)

- **Enshan C**, Zihan X, Anan T, Ziyue Y, Yuqing Z, & Zheng Z. (2024). 'Book mountains veiled in shadows,' Shanghai Jinshan Library design competition, Third prize
- **Enshan C**, Yuchuan K. (2014). 'Nature Courtyards,' Chongqing University Lixin Building design competition (Student group), First prize
- **Enshan C**. (2013). 'Cliff rescaping,' Yuandao Cup college student design competition, Second prize

(Working period)

- Suyang L, **Enshan C**, Nanli. (2017). Chongqing Banan Babinglu commercial complex design competition.
- Weixi N, **Enshan C**, etc. (2017). Sany Group Hainan residential project competition.
- Tao Z, **Enshan C**, Cheng S, Da L. (2016). Taiyuan Yangjiayu Macalline residential district design competition.
- Qiuhaol L, Yapeng W, **Enshan C**, Chuanqi S. (2016). Beijing Meitougou Longfor commercial complex (including a shopping mall, office towers, and residential towers) design competition.

Exemplary design-related content

- Design competition of Jinshan library, Sep 2024:





- The **design and research portfolio during the Master's degree time, 2018-2021**, (30+ pages, [click the link](#)):

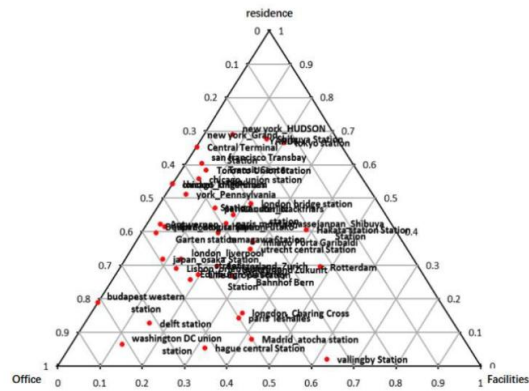


Figure 6: The MXI graph based on the data collected

General pattern and cases of function
The American cases contain a high proportion of office functions, which is consistent with the phenomenon that American trains are mostly used for short-distance office commuting. Europe is represented by countries such as the Netherlands, with a high degree of function mix, and stations both as transportation nodes and as places of activities are strongly reflected (Figures 10 and 11).



Figure 11: The east side of the utrecht central station area



Figure 12: The west side of the utrecht central station area

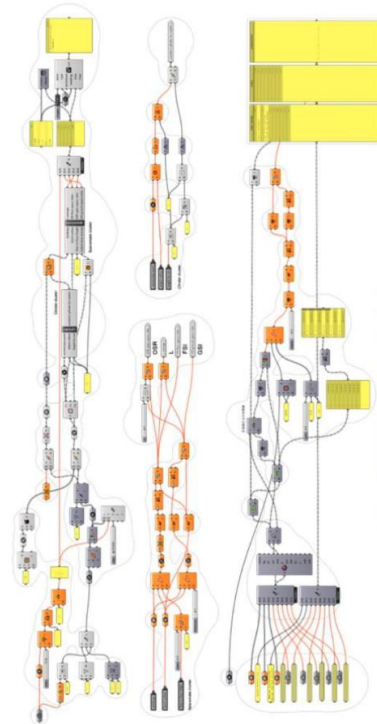
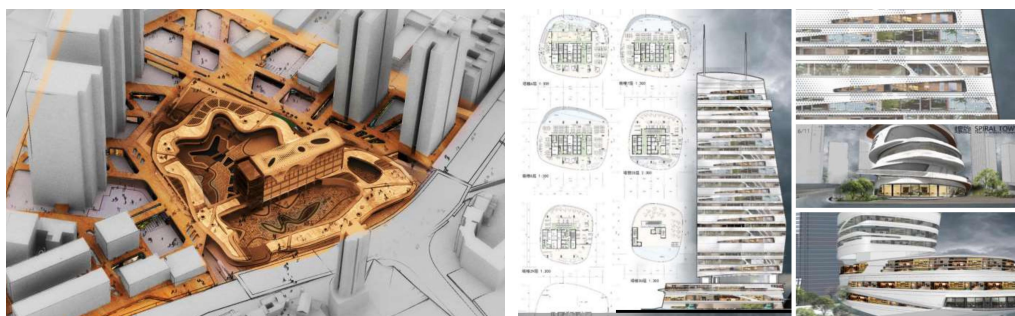


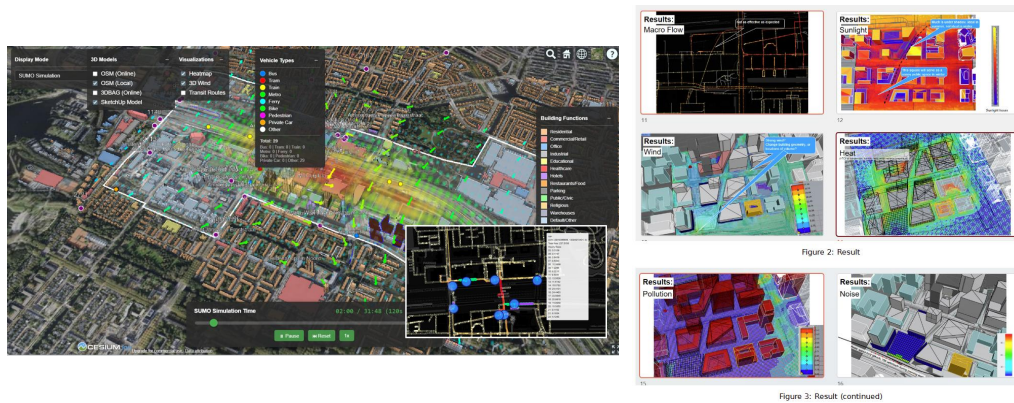
Figure 13: Extract the density and function data by GH

- The **design portfolio before and during working at Shanghai Tianhua, 2015-2017**, (30+ pages, [click the link](#))
- Parametric design during bachelor's degree time, 2013-2014:



7 Demos of my digital works

A digital twin for Assessing Multisensory User Experience (AMUSE) (github.com/1309928130/MultSensory-Digital-Twins)



A macroscopic pedestrian simulation model (gitlab.tudelft.nl/xcarcity/digital-twin/en-shan/macroscopic-pedestrian-model)

Selected Web Applications & Utility Tools:

- Dutch sentence learning web app (dutch-sentence-trainer.web.app)
- Quantitative assessment preparation platform (optiver-prep.web.app)
- Screen scrolling capture tool (github.com/1309928130/TOOLS_scroll-capture)
- Batch images to PDF converter (github.com/1309928130/TOOLS_images-to-pdf)

8 Publications

Published journal articles (Peer-reviewed)

[During PhD time]

Enshan, C., van de Spek, S., van der Hoeven, F., & Triggianese, M. (2025). Evaluate user satisfaction for urban design of railway station areas: An assessment framework using agent-based simulation. *Environmental Impact Assessment Review*, 110, 107685. <https://doi.org/10.1016/j.eiar.2024.107685> [Journal quartile: Q1; Impact factor: 11.2; SSCI]

Enshan, C., van der Spek, S., Tian, A., Triggianese, M., & van der Hoeven, F. (2025). Evaluate design principles for flexible use of railway station areas: Through research-by-design and agent-based simulation. *Journal of Building Engineering*, 112, 113775. <https://doi.org/10.1016/j.jobbe.2025.113775> [Journal quartile: Q1; Impact factor: 7.4; SCIE]

Chen, E., & Zhuang, Y. (2022). The Stakeholder Gaming and Element Organization—Research on the Station-city Integration Development of Utrecht Central Railway Station [In Chinese]. *The Architect*. 3, 52–60. <https://doi.org/10.12285/jzs.20211015002>

[During master's degree time]

Chen, E., Jiang, M., & Zhuang, Y. (2021). Research on Urban Morphology Characteristics of Railway Station Based on Density and Function Index [In Chinese]. *Architecture Technique*, 27(04), 100–102. <https://doi.org/10.19953/j.at.2021.04.024>

Chen, E., Yang, S., & Zhuang, Y. (2021). Analysis of Urban Security Based on Visibility: Taking Utrecht Railway Station Area as an Example [In Chinese]. *Architecture Technique*, 27(04), 121–123. <https://doi.org/10.19953/j.at.2021.04.031>

Enshan, C., & Wenda, S. (2021). Analysis of Spatial Features of Adjacent Neighborhood and Preliminary Discussion of Renew -Taking Four Adjacent Neighbors of Jiangxibei Road and Wujin Road of Shanghai City as an Example [In Chinese]. *Housing Science*. <https://doi.org/10.13626/j.cnki.hs.2021.01.007>

Published conference papers, posters, or presentations (Peer-reviewed)

Zhoulangyi Xing, Junting LIN, **Enshan CHEN**. (2025) Exploring Functional Configuration in European Station Areas Based on Multi-Scale Accessibility Quantification. **AESOP ANNUAL CONGRESS 2025**

Anan Tian, **Enshan Chen** (2025). Mediated Architectural Repositories: Performing a Context-Driven, Spatio-Temporal VR Archive Prototype. CDRF 2025 (The 7th International Conference on Computational Design and Robotic Fabrication) [**Digital futures** 2025]

Enshan, C., van der Spek, S., van der Hoeven, F., & Triggianese, M. (2024). Study the Design Principles for Fluctuation-Responsive Railway Station Areas. The 16th Conference of the International Forum on Urbanism (**IFoU**), Multi-city Online. https://www.researchgate.net/publication/384561119_Study_the_Design_Principles_for_Fluctuation-Responsive_Railway_Station_Areas [Abstract + Full paper]

Enshan, C., Stefan, van der S., Frank, van der H., & Manuela, T. (2024). Assessing user satisfaction for urban design of railway station areas. **ACSP** 64th Annual Conference, 1257-1258. https://cdn.ymaws.com/www.acsp.org/resource/resmgr/2024_conference/docs/acsp2024_bk_of_abstracts.pdf [Abstract]

Chen, E., van der Spek, S. C., van der Hoeven, F. D., & Triggianese, M. (2022). Station City Integration in China: Towards Mobility Resilience and Public Space Flexibility. **Urban Transitions 2022: Integrating Urban and Transport Planning, Environment and Health for Healthier Urban Living**. https://www.researchgate.net/publication/381189110_Station_City_Integration_in_China_Towards_Mobility_Resilience_and_Public_Space_Flexibility [Poster]

Published book chapters

Zhuang, Y., & **Chen, E.** (2022). The urban form of the railway station area. In *Station-city integration: Urban design* (In Chinese), 191-226. China Architecture & Building Press.

Journal articles in submission or mostly finished

Enshan, C., Jie, G., & Snelder, M. Computational design assessment regarding multisensory user experience: A workflow 【Postdoc project work】

Enshan, C., van der Star, G., Nadi, A., Duvies, D., Damen, W., & Snelder, M. Towards Walkable Cities: A High-Resolution Spatiotemporal Macroscopic Pedestrian Modeling Framework Using Open Data for Urban Design in Dutch Cities. 【Postdoc project work】

Enshan, C., van de Spek, S., van der Hoeven, F., & Triggianese, Tian, A., M. Lin, J. Problem statement of the overcrowding and emptiness in railway station areas: Case studies 【PhD project work package 1-a】

Enshan, C., van de Spek, S., van der Hoeven, F., & Triggianese, M., Cui, M. Inquiring fluctuation-supportive spatial solutions addressing overcrowding and emptiness in railway station areas: A literature review 【PhD project work package 1-b】

Enshan, C., van de Spek, S., van der Hoeven, F., & Triggianese, M. Investigating the design princi-

ples for fluctuation-supportive Railway Station Areas [【PhD project work package 3; Click this link to see the project website】](#)

Enshan, C., van de Spek, S., van der Hoeven, F., & Triggianese, M. Anan, T., Junting, L., Lanyi, X. Assessing visual experience: Evidence extracted from image segmentation

9 Academic / teaching activities

Sep 2026. To give a lecture in the Master's course MADE at the Amsterdam Institute for Advanced Metropolitan Solutions.

Jan-May 2026. Served as the daily supervisor in Bachelor's End Project (Bachelor's graduating thesis), for a group of 6 students, at the Transport Planning Department, Civil Engineering Faculty.

17-21/03/2025. Served as a critic in an international workshop: [Reimagining Architecture of Interchange](#). For a group of 35 students in the [ATHENS Network](#). TU Delft, the Netherlands

03/2025. (Online) Co-tutoring in a course: How to expand your discipline and keep it relevant in transformative times. Tongji University, Shanghai, China.

25/09/2024. Organized Workshop: [Fluctuation-responsive railway station](#). TU Delft, the Netherlands

22/06/2024. Present Lecture: Station-City Integration, towards the Dynamic Model: The problems, assessment, and guidelines. Active mode lab, faculty of Civil Engineering and Geosciences, TU Delft, the Netherlands

15/04/2024. Participant Serious Game: Crossing the Farm (a board game testing how the food system might form a future self-sustainable city, with the participation of multiple stakeholders). Rotterdam, the Netherlands

12/03/2024. Participant Workshop: Low carbon stations. TU Delft, the Netherlands

16/11/2023. Attend Conference: National Stations Congress. 's-Hertogenbosch, the Netherlands

10 A brief research plan for the coming years

My research interests lie in the urban and architectural design of complex systems, meanwhile supporting design with emerging technologies. I built two frameworks linking rich content and technologies, as Figs. 2 and 1. During my PhD time, I have delved into some parts of the complex systems (colored texts in Fig. 2), and mainly focused on simulation as a technical method; In the future, my vision is to lead a lab to continue exploring other components in these two frameworks.

Future research spectrum

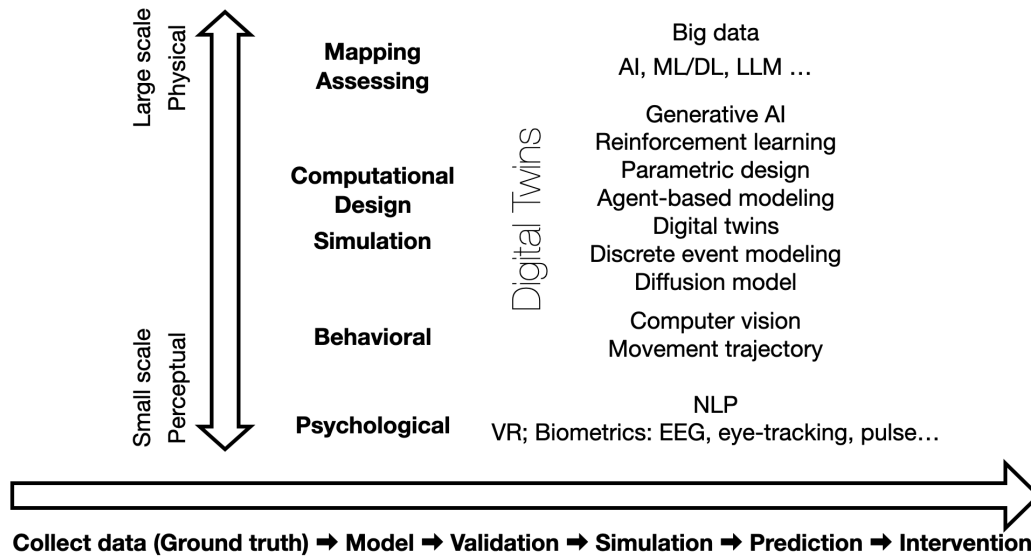


Figure 1: Research spectrum regarding technologies

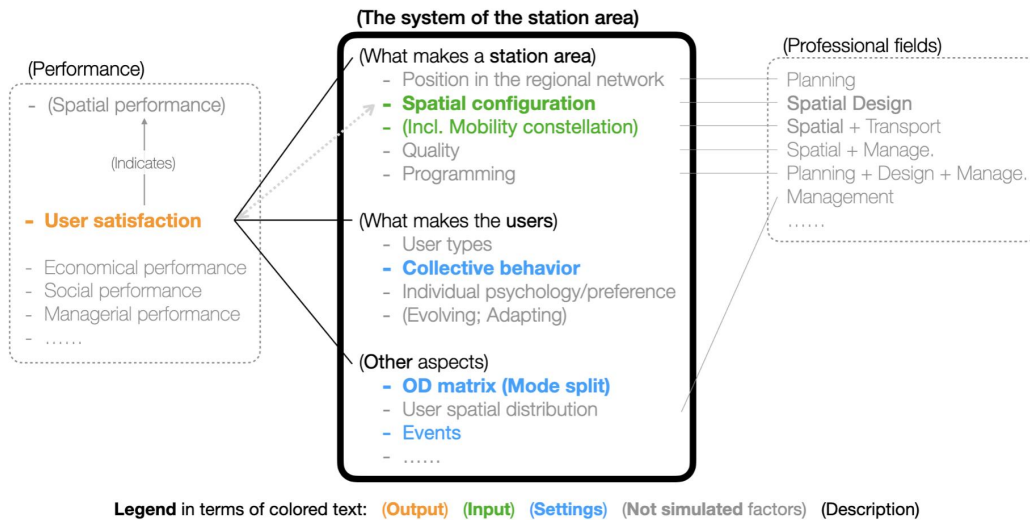


Figure 2: Simulating the system of complex projects, taking railway stations as an example

11 Contact details for referees

Postdoc supervisors:

- Associate Prof. Dr. **Maike Snelder**, M.Snelder@tudelft.nl, +31 15 27 84981
- Assistant Prof. Dr. **Jie Gao**, J.Gao-1@tudelft.nl
- Research Scientist Dr. **Ali Nadi**, Ali.Nadi@tno.nl

PhD supervisors

- Associate Prof. Dr. **Stefan van der Spek**, S.C.vanderSpek@tudelft.nl, +31 15 27 89860
- Associate Prof. Dr. **Manuela Triggianese**, M.Triggianese-1@tudelft.nl, +31 15 27 87468
- Associate Prof. Dr. **Frank van der Hoeven**, FDvanderHoeven@tudelft.nl, +31 15 27 88462

Master's supervisor

- Prof. Dr. **Yu Zhuang**, arch-urban@163.com, +86 18918221087